

Lecture 7. Hypothesis Testing

In psychological research, we often want to know whether one variable (IV) really does affect another (DV), or whether two variables really are associated. Unfortunately, statistical testing can't tell us whether there are such effects or relationships in the real world, because our tests do not possess knowledge about the real world. Instead, we can infer whether a true effect is *likely* or *unlikely* to exist based on a sample that we've taken from the real world:

1. Imagine how the world would look if there was *no effect* or *no relationship* – we call this the *null hypothesis*.
2. Collect some data from our world and describe it.
3. Calculate how *likely* it would be to get data that looks like this if we lived in a world where there was no effect/relationship.
4. If the likelihood is very low (i.e., if it's highly unlikely to find data like ours provided no effect/relationship exists), then we can be highly confident that there is, in fact, an effect/relationship.

Think about it

Notice two interesting implications of this approach:

(1) As researchers, we should not be focused on trying to prove that we are right (e.g., that an effect exists). Instead, we should try to prove that we are wrong (i.e., try to find evidence that there is *no effect*). If we *fail* in this, we can be confident that in reality there is an effect. In practice, this means that we should not be trying to prove that our experimental hypothesis is true, but instead **try to prove the null hypothesis ("There is no effect") – and fail to do so**.

(2) We can never say conclusively that we have found an effect in a single study. The best we can do is to say that it would be highly unlikely to obtain data that looks like ours if there was no effect – but a slight possibility always remains.

Step 1. Imagine how the world would look if there was no effect or no relationship.

When we make a prediction about the world, we express it in the form of a hypothesis. We can further distinguish between:

- the more abstract **research hypothesis** that describes the predicted effect (e.g., "I can predict the future"), and
- the more concrete, testable **statistical hypotheses** that are mathematically precise and correspond to specific outcomes in the study. For each study, we need to have at least two such hypotheses describing two alternative outcomes:
 - **Null hypothesis (H_0)**: Expected result if there was no effect.
 - **Alternative hypothesis (H_1)**: Expected result if there is an effect.

Example

- **Research hypothesis:** *I can predict the future*
- **Statistical hypotheses:**
 - **Null hypothesis (H_0):** number of correct responses is no different to what we would expect by chance if I were guessing
 H_0 : Correct responses = 10
 - **Alternative hypothesis (H_1):** number of correct responses is greater than or less than we would expect by chance H_1 :
Correct responses $\neq$ 10

Step 2. Collect some data and describe it

I can do 20 coin flips, predicting the outcome of each one, and recording the number of correct predictions.

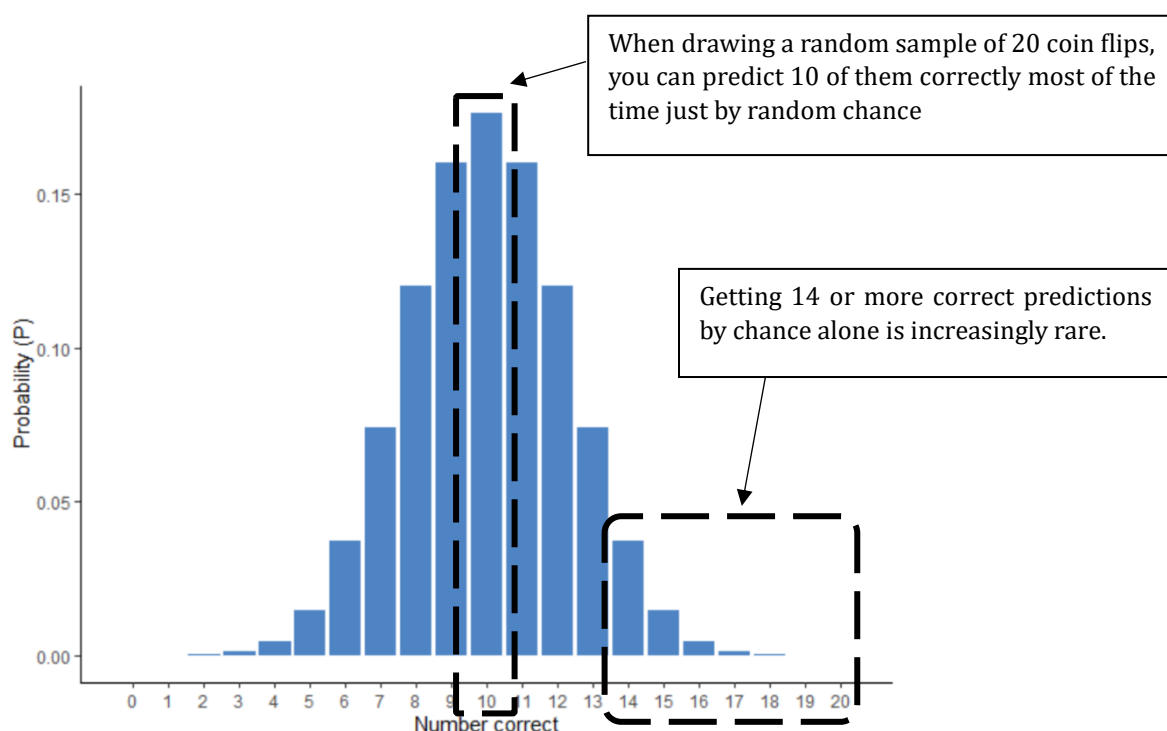
I also need to choose a **test statistic** that describes my data – in this case, the test statistic is the number of correctly predicted coin flips (in the case of my experiment – 14). Does this number of correct predictions convince you to reject the null hypothesis?

Step 3. Calculate the likelihood of observing this data if H_0 is true

We know that, if H_0 is true (“I *can’t* predict the future”), then we should be predicting about half of all coin flips (10/20) correctly; however, we also know that we need a large enough sample to detect that rate accurately – small samples can produce inaccurate results due to statistical noise. Therefore, if H_0 is true, one could still make 14 correct predictions due to random chance. But how *likely* is it?

Here we can deploy the skills we learned from our lecture on sampling. We can either use the **theoretical probability** (use reasoning to calculate the likelihood of getting 14/20 flips correctly by random chance) or simulate data to calculate the **empirical probability** (e.g., use R to randomly pick “correct” or “incorrect” 20 times, and repeat the same trial over many, many iterations). You can try doing that yourself – the code is in the “Bonus activity” section at the end of these notes.

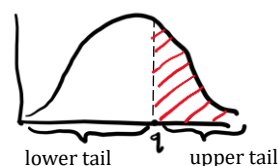
We can use a histogram like the one below to visualise the distribution of correct predictions assuming that H_0 is true. Each bar represents how many of our random samples contained a given number of “correct” predictions. Since H_0 is true in this simulation, most samples should have produced around 10 correct predictions out of 20, but some of them will have more/fewer correct predictions by chance. The more extreme the results, the less likely they should be.



Now that we have the distribution of samples given $H_0 = \text{True}$, we can use our knowledge of probabilities to calculate the likelihood of observing a result as extreme as ours (14/20 correct predictions) by chance alone. Remember the `pbinom()` function from the lecture on probabilities:

```
pbinom(q = 13,          # Calculate the likelihood of getting >13 correct
       size = 20,       # out of 20 attempts
       prob = .5,       # where each attempt has a 50% chance of correct prediction
       lower.tail = FALSE)
```

Note: `pbinom()` creates a distribution of scores divided by the value q (in this example, we've set $q=13$). Anything below q is the lower tail of the distribution. We need to specify `lower.tail=FALSE` in order to *exclude* this range (0-13). What we're left with is the upper tail of the distribution (above q), i.e., 14-20.



By running the code above, we get a probability value (**p-value**) of 0.057 – i.e., if someone tried to predict the results of 20 coin flips by chance alone, they would get 14 or more of them right only 5.7% of the time.

Notice how the p-value does not tell us whether this outcome *is* the result of random chance. It merely tells us how (un)likely the outcome is if only random chance is at play. From this point on, we have to make an active judgement on whether to retain or reject H_0 based on this probability.

Step 4. Decide if the outcome is too unlikely to be the result of chance

In psychology, it has become conventional to use the 5% threshold for deciding whether a finding is so unlikely that we can reject H_0 . That means that getting a p-value of .05 or lower is usually taken as evidence that there is a real effect.

Notice that this can be counter-intuitive: **we want small p-values**, as we want results that are unlikely to arise randomly if there was no effect.

However, you should also keep in mind that the $p < .05$ threshold is merely conventional – it has become established in research practice because it is convenient to calculate, not because anything magical happens around that level of probability. If we get $p = .05$, that means that results as extreme as ours can still happen by chance about 5% of the time. Even with $p = .01$, we would expect to see such extreme results in 1 trial out of every 100 (1%) – given how many studies are run each year, this could still produce a huge number of findings that look like real effects but are in reality statistical noise.

Type 1 and Type 2 Error

There are four possible outcomes to our study:

		Study outcome	
		I retain H_0 (I conclude I <i>can't</i> predict the future)	I reject H_0 in favour of H_1 (I conclude I <i>can</i> predict the future)
Truth	H_0 is true (I <i>can't</i> really predict the future)		Type I error: false positive (I believe I've found an effect where there is none)
	H_0 is false (I <i>can</i> really predict the future)	Type II error: false negative (I fail to detect a true effect)	

We need to be aware of the error rates:

- **Type I (false positive) error depends on the threshold** we've set for rejecting H_0 . We express this as an **α (alpha) level**. In the previous section we mentioned that the standard threshold is 5%, i.e., $\alpha = .05$. That means we will reject H_0 and accept H_1 when we get results with a probability (p-value) of .05 or less. When $\alpha = .05$, we accept a 5% risk of a Type I error.

- **Type II (false negative) error depends on statistical power.** If our study is under-powered (e.g., due to a small sample size), we may end up failing to detect a true effect. The Type II error rate is expressed as **beta (β)**. The power of our test is $1-\beta$. For example:
 - $\beta=.20$ means that we will fail to reject a false H_0 (i.e., fail to detect a true effect) 20% of the time.
 - The power of the test we're running will be $1-\beta = .80$, which means that the test has the power to detect a true effect about 80% of the time.

In order to produce reliable results, we need to minimise the chance of Type I and II errors. Lowering the α threshold means we'll detect false effects less often. Lowering the β level means we'll have sufficient power to detect true effects more often.

How to interpret p-values

Some people (e.g., statistician Ronald Fisher) advocate using the p-value as a hard boundary of statistical significance, allowing us to make a categorical judgement:

- if $p < 0.5$, we have evidence against H_0 . Results at $p < .05$ are "statistically significant"
- if $p > .05$ we have no evidence against H_0 . Results at $p > .05$ are "not significant"

Others (like J. Neyman and E. Pearson) have advised against making binary judgements like this. Instead, they say the p-value should be the smallest error rate (α) you would have to be willing to tolerate in order to reject the null hypothesis – while remembering that the possibility of error is still present.

p-values are not...

- ✗ *The probability of H_0 being true.*
- ✗ *The probability you're making the wrong decision.*
- ✗ *The probability that you would get the same result if you ran the study again.*
- ✔ Actually, the p-value is the probability of *observing a test statistic* (or one more extreme) if the null hypothesis were true. In other words, p-values are not probabilities of hypotheses, they are probabilities of the outcomes.
- ✗ *The probability that you have found a meaningful effect.*
- ✔ In statistics, "significant" does not mean "meaningful". A result can be small (practically meaningless) but still statistically significant. In other words, we can have statistical evidence for very small effects that are not actually meaningful in practice. To know whether a result is practically meaningful, we need to look not at the p-value, but the **effect size**.

The problem of running multiple tests

$\alpha=.05$ means that, when the null hypothesis is true, we will still get a significant result by chance 5% of the time. If we test 20 hypotheses with an alpha level of .05, then we would expect at least one of these to come out as significant just by chance.

Therefore, if you go into a dataset and test multiple hypotheses to “find something interesting”, you are likely to be finding something that is a spurious result!

In order to control for this, we can set an α level that is much more stringent than .05 – to keep a 5% error rate across all 20 tests, we would need to have an α of $0.05/20 = .0025$ instead. That means that we would accept any of these 20 results to be statistically significant only if $p < .0025$. This is called a **Bonferroni correction** and should be applied whenever we make multiple unplanned comparisons.

Key terms to remember

H₀ and H₁: Null hypothesis and alternative (experimental) hypothesis.

Test statistic: a statistical value that expresses our data (e.g., t, r, F, χ^2 – we’ll learn more about these in the coming weeks).

Effect size: the strength of the observed effect, e.g. the size of the difference between two groups, or the strength of the association between two variables.

p-value: the probability of observing this particular value of our test statistic assuming that H₀ is true.

α (alpha level): the threshold that the p-value must reach for us to reject H₀. This is also the **Type I** (false positive) error rate. When $\alpha=.05$, we need to observe $p < .05$ to declare the results statistically significant, and we also have a 5% risk of observing such a result by chance alone (i.e., 5% risk of a false positive result).

β (beta level): the Type II error rate (false negative = failure to detect a true effect).

Statistical power: the test’s capacity to detect a true effect. Power = $1 - \beta$.

Bonferroni correction: a correction we must make to our critical threshold (α) to account for multiple comparisons. The corrected $\alpha = \frac{.05}{\text{Number of comparisons}}$.

Bonus activity

Let's try to replicate our coin-flipping experiment. We'll generate 500 samples of 20 coin flips each, where the chance of correctly predicting each flip is 50%. We'll then plot the resulting trials in a histogram and observe their distribution.

```
# Load the tidyverse
library(tidyverse)

# Create some sample data - 10,000 trials in which we predict
the outcome of the coin flip either correctly or incorrectly
statements <- sample(c("correct", "incorrect"), 10000, replace =
TRUE)

# Now we can start sampling from this data. Make a blank list
where our 500 samples will go. If you want to try this with
more/fewer samples, just change the number.
results <- numeric(500)

# The following code runs a loop 500 times. Each time it takes a
random sample of 20 trials that resulted in "correct" or
"incorrect" predictions; counts the numbers of "correct"; then
saves that number in the object "results". It repeats this
process until it gets all 500 samples.
for (i in 1:num_trials) {
  sampled_statements <- sample(statements, 20)
  num_correct <- sum(sampled_statements == "correct")
  results[i] <- num_correct
}

# Save the results of our 500 samples as a data frame called
"data", which then can be plotted with ggplot
data <- data.frame(results)

# Visualise the number of correct predictions across our 500
samples using a histogram.
ggplot(data, aes(x = results)) +
  geom_histogram(binwidth = 1, fill = "white", color = "black")
+
  labs(
    x = "Number of Correct Statements Sampled",
    y = "Frequency",
    title = "Distribution of Correct Statements in 500 Trials"
  )
```